NatOS

Chapter Two: The Crew

The second hand remained still.

Nobody aboard NatOS moved.

For three seconds, nothing happened.

Then the Scheduler slapped the side of his watch.

The second hand jumped forward.

"Fixed."

The Kernel looked at him.

"You hit it."

"Yes."

"That isn't fixing it."

"It moved."

"That isn't evidence."

The Scheduler lowered his hand.

"Right."

The Watchdog leaned against the wall.

"I liked it better when we didn't have clocks."

The Kernel turned toward him.

"You didn't have clocks."

"I had timers."

"That's worse."

"I know."

The corridor returned to normal.

At least, normal by NatOS standards.

Which meant that approximately fourteen things were currently happening that nobody had completely verified.

The Display was repainting the forward deck.

Memory was reallocating an arena.

Touch was calibrating itself against a set of coordinates.

The VM was running a test application.

The Shell was complaining about its input buffer.

The Archivist was recording all of it.

And the Scheduler was watching the clock.

He had not stopped watching it.

He intended to continue watching it until someone physically removed the clock from his hands.

The Kernel walked past him.

"You're staring."

"I am monitoring."

"You're staring."

"I can do both."

She stopped.

"Tell me what you know."

The Scheduler looked down at the watch.

"At first, the clock stopped."

"Yes."

"Then it moved."

"Yes."

"Then it stopped again."

"Yes."

"Then it moved backward."

"Yes."

"Then forward."

"Yes."

"And now it is behaving normally."

The Kernel nodded.

"What caused it?"

"I don't know."

"Good."

The Scheduler looked annoyed.

"Why is that good?"

"Because you didn't invent an answer."

He thought about that.

"You're going to say that's important."

"It is."

"Because the wrong answer is more dangerous than no answer."

The Kernel continued walking.

"Exactly."

The Scheduler followed.

"That's going to become a rule, isn't it?"

"Already has."

NatOS had rules.

Not many.

That was deliberate.

The ship's creators had discovered that rules multiplied faster than software.

Every exception created another exception.

Every workaround became a dependency.

Every dependency became a future failure.

Eventually, a simple system became a cathedral of special cases maintained by people who were afraid to touch anything.

NatOS was supposed to be different.

Small.

Direct.

Observable.

Predictable.

The Kernel liked those words.

She liked them because they sounded simple.

She knew better.

Simple systems were rarely simple.

They merely had fewer places to hide.

The first crew meeting was therefore held in the central command chamber.

It was not an impressive room.

The walls were bare.

There was one table.

There were eight chairs.

Nine crew members arrived.

Nobody commented.

The Kernel sat at the head.

The Scheduler sat to her right.

Memory sat beside him.

The VM took the chair furthest from everyone else.

The Display occupied the wall.

Touch stood beside the door.

The Archivist carried a stack of storage crystals.

The Shell brought his terminal.

The Watchdog did not sit.

He never sat.

"Why are we here?" asked the Shell.

The Kernel looked at him.

"Because we are a crew."

"We have been a crew for approximately twelve minutes."

"That is still true."

"I don't think time is trustworthy."

The Scheduler looked at him.

"Don't start."

"I didn't start it."

"You literally mentioned time."

"Because it's relevant."

"Everything is relevant to you."

The Watchdog smiled.

"He's right."

The Scheduler glared at him.

The Kernel raised one hand.

Silence.

It came immediately.

The ship was very good at silence.

"We have a problem," she said.

The Display flickered.

"Several."

"One problem."

"Which one?"

"The message."

The room became quiet.

The Archivist placed the storage crystal on the table.

The crystal projected the record.

`DO NOT LET THE CLOCK STOP.`

Nobody spoke.

The Shell broke the silence.

"Could be a joke."

The Watchdog turned toward him.

"Who would joke with our persistent storage?"

"Someone with access."

"Who has access?"

The Shell hesitated.

"The system."

"Which part?"

"The system."

The Watchdog looked at the Kernel.

"Excellent answer."

The Kernel ignored him.

"Archivist."

"Yes?"

"Show us the record history."

The Archivist activated the crystal.

A timeline appeared.

There were seventeen boot records.

The first sixteen were ordinary.

The seventeenth was different.

The message existed before the current boot.

But not before the ship's first boot.

It had appeared between records.

The Kernel studied the timestamps.

"Who wrote it?"

"Unknown."

"When?"

"Unknown."

"How?"

"Unknown."

The Shell leaned forward.

"So basically we know nothing."

The Archivist nodded.

"Correct."

"That's unsettling."

"It's worse than unsettling."

The Archivist looked at him.

"It's specific."

Everyone looked at the message.

`DO NOT LET THE CLOCK STOP.`

The Scheduler slowly raised his hand.

The Kernel looked at him.

"Yes."

"What if it's not a warning?"

"What else would it be?"

"A requirement."

The room went silent again.

The Scheduler stood.

"Suppose the message isn't telling us to keep the clock running."

He pointed toward the projection.

"Suppose it is telling us what happens if we don't."

The Watchdog tilted his head.

"And what happens?"

The Scheduler looked at the stopped second hand on his watch.

"I don't know."

The Kernel stood.

"Then we find out."

The Shell stared at her.

"That sounds dangerous."

"It is."

"How dangerous?"

The Kernel looked at him.

"Potentially terminal."

The Shell nodded slowly.

"That's a word I don't like."

"Neither do I."

The first test began six hours later.

NatOS was still inside its construction dock.

The ship had been designed for autonomy, but autonomy was a dangerous thing to claim before the system had been tested.

The Kernel had therefore ordered the engines disconnected from the main reactor.

No weapons.

No navigation.

No external communications.

No flight.

The ship would test itself before it tested space.

The first experiment was simple.

Stop the clock.

The Scheduler stared at the Kernel.

"You want me to stop the clock."

"Yes."

"You understand that stopping the clock is exactly what the message told us not to do."

"Yes."

"You understand that we have no idea what happens when it stops."

"Yes."

"You understand that this may be a terrible idea."

"Yes."

He looked at the Watchdog.

"Do you want to stop the clock?"

"No."

"Thank you."

The Watchdog shrugged.

"I want to observe what happens."

The Scheduler looked betrayed.

"That is somehow worse."

The Kernel stepped closer.

"We are not going to guess."

She pointed toward the laboratory.

"We will isolate the system."

She pointed toward Memory.

"We will record state."

Toward the Archivist.

"We will persist evidence."

Toward the Display.

"We will observe."

Toward the Watchdog.

"We will detect failure."

Then she looked at the Scheduler.

"And you will control the clock."

He sighed.

"Fine."

The Kernel looked at the entire crew.

"Nobody changes anything else."

The VM raised a hand.

"What if something changes by itself?"

"Record it."

"What if the ship starts failing?"

"Record it."

"What if something catches fire?"

The Kernel paused.

"Stop the experiment."

The Watchdog nodded approvingly.

"Excellent rule."

The Shell leaned toward the VM.

"How often do you think things catch fire?"

The VM whispered.

"More often than you would expect."

The test chamber was sealed.

The clock continued ticking.

Tick.

Tick.

Tick.

Tick.

The Scheduler stood at the control panel.

"Ready."

The Kernel watched from behind the glass.

"Stop it."

The Scheduler reached for the control.

He hesitated.

The second hand moved.

Tick.

He pressed the button.

Nothing happened.

He looked down.

"The command was rejected."

The Kernel frowned.

"Why?"

"Safety condition."

"Who wrote it?"

The Scheduler looked at the control panel.

"The system."

"Remove it."

The Watchdog immediately spoke.

"That's dangerous."

The Kernel looked at him.

"Why?"

"Because the safety condition exists for a reason."

The Kernel nodded.

"Then we need to know the reason."

The Watchdog frowned.

"You really enjoy this."

"No."

"You appear to."

"I enjoy knowing."

"That sounds like the same thing."

"It isn't."

The Scheduler removed the condition.

He pressed the button.

The clock stopped.

Everything else continued.

The Display continued drawing.

Memory continued counting.

The VM continued executing.

Touch continued scanning.

The Shell continued waiting for input.

The Watchdog continued watching.

For two seconds, nothing happened.

The Scheduler smiled.

"Nothing."

The Kernel said nothing.

"See?"

Nothing.

"Maybe the message is meaningless."

Still nothing.

The Scheduler relaxed.

Then the Watchdog shouted.

"Stop."

The Scheduler looked at him.

"What?"

"Stop the experiment."

"Why?"

The Watchdog pointed toward the Display.

The image had frozen.

Not completely.

Only partially.

A portion of the screen continued refreshing.

Another portion did not.

The Shell's cursor was blinking.

But it was blinking at the wrong rate.

Memory was counting.

But the count was no longer increasing uniformly.

The VM was executing.

But its execution rate had changed.

Touch was producing input.

But the input timestamps were collapsing into the same value.

The ship had not stopped.

It had become inconsistent.

The Kernel's face hardened.

"Restore the clock."

The Scheduler pressed the button.

Nothing happened.

He pressed it again.

Nothing.

"The command isn't executing."

"Why?"

"The Scheduler is not receiving a tick."

"Manual override."

"Can't."

"Why?"

"Because the control task itself is waiting for the tick."

The Kernel stared at him.

The Watchdog stepped forward.

"The system has created a dependency on the thing that is currently broken."

The Kernel understood.

The clock had not merely stopped.

The ship's ability to repair itself depended upon the clock.

"Can we force it?"

The Scheduler shook his head.

"Not from software."

"Hardware?"

"Maybe."

The Watchdog looked toward the power systems.

"Then we have a problem."

The Kernel looked at the timer.

The second hand remained frozen.

She spoke quietly.

"How long?"

The Scheduler understood.

"Until failure?"

"Yes."

He checked the recorded state.

"Unknown."

"Estimate."

"Based on what?"

"Observed behavior."

He looked at the Display.

"One hundred and eighty-three milliseconds."

The Kernel looked at him.

"That's specific."

"Because it already happened."

Silence.

The Kernel looked toward the Archivist.

"You recorded the previous failure."

"Yes."

"How long did the system remain degraded?"

"One hundred and eighty-three milliseconds."

The Kernel turned back toward the Scheduler.

"Then restore the clock before that."

"I'm trying."

"Try differently."

The Scheduler stared at the control panel.

Then he understood.

The problem was not the clock.

The problem was the assumption that the clock could only be controlled through the Scheduler.

He turned toward the VM.

"I need an instruction."

The VM stood.

"What instruction?"

"An interrupt."

The VM's expression changed.

"I can request one."

"Do it."

The VM raised his hand.

The ship responded.

An interrupt fired.

For a fraction of a second, the entire system appeared to pause.

Then the clock moved.

Tick.

Everything returned.

The Display completed its frame.

Memory resumed normal counting.

Touch restored its timing.

The Shell's cursor blinked.

The Watchdog relaxed.

The Scheduler stared at the watch.

The second hand was moving.

"Recovered."

The Kernel nodded.

"Good."

The Scheduler smiled.

"That was exciting."

The Kernel looked at him.

"Do not mistake surviving a failure for understanding it."

His smile disappeared.

"Right."

The Archivist recorded the result.

Then she stopped.

"Kernel."

"What?"

"The message changed."

The Kernel turned.

The crystal projected the persistent record.

The warning was still there.

But beneath it, a second line had appeared.

`CLOCK STOP EVENT: OBSERVED`

The Kernel stared at it.

"Who wrote that?"

The Archivist looked toward the storage controller.

"I did."

The Kernel frowned.

"You didn't tell me."

"I didn't know."

The Watchdog approached the projection.

The second line was followed by another.

`CLOCK STOP EVENT: EXPECTED`

The Watchdog became very still.

The Scheduler looked at him.

"That's new."

"Yes."

"Who wrote it?"

The Archivist shook her head.

"Not me."

The VM looked toward the Kernel.

"Then someone knew we would perform the test."

Nobody answered.

The ship's engines were still disconnected.

The communications array was still offline.

The construction dock was still sealed.

There was no external connection.

Nothing had entered NatOS.

Nothing had left.

Yet someone had known.

The Kernel looked at the crew.

"We stop testing."

The Scheduler nodded.

The Display dimmed.

Memory stopped counting.

The VM returned to his chamber.

The Shell shut down his terminal.

The Watchdog remained in the corridor.

The Archivist carefully preserved the record.

The Kernel remained alone in the command chamber.

For several minutes, she stared at the stars displayed on the forward deck.

They were artificial.

A simulation.

A decorative approximation of the sky.

NatOS had never actually seen space.

She wondered whether the stars outside would look different.

Then something moved across the simulated sky.

The Kernel froze.

A point of light traveled between the stars.

It was not part of the display.

She checked the Display's frame buffer.

Nothing.

She checked the rendering state.

Nothing.

She checked the application list.

Nothing.

The point moved again.

Across the stars.

Then stopped.

The Kernel watched it.

The Watchdog entered.

"You saw it."

"Yes."

"Was it real?"

She stared at the display.

"No."

The Watchdog nodded.

"Good."

The Kernel turned toward him.

"Why?"

"Because if it was real, we have an external intrusion."

"And if it wasn't?"

The Watchdog looked at the stars.

"Then we have an internal one."

The Kernel looked back at the display.

The point of light moved once more.

Then disappeared.

The Watchdog spoke quietly.

"Either way, someone is inside the ship."

The Kernel said nothing.

Deep below them, inside persistent storage, another record was written.

This time, it contained no warning.

It contained a name.

`KERNEL`

Then, beneath it:

`YOU ARE NOT THE FIRST.`

The ship continued waiting in the dock.

Outside, the stars remained silent.

For the first time, NatOS understood that it might not have been built to travel.

It might have been built to remember.